



Why the confusion matrix fails as a model of knowledge

Ian van der Linde¹

Received: 18 February 2025 / Accepted: 21 June 2025 / Published online: 2 July 2025
© The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature 2025

We tend to think that if something is true, and we have good reasons to believe that it is true, then we “know” it. This idea, described by Plato as far back as 360 BCE, is known as the Justified True Belief (JTB) model of knowledge. To “know” something, you have to (1) believe it, (2) it has to be true, and (3) you need good reasons or evidence to justify that belief. For instance, if I look out of the window and see rain and also hear rain hitting the roof, I might say I “know” it is raining. This is knowledge, because (1) I believe it, (2) it truly is raining, and (3) I have good evidence to support my belief (both seeing and hearing the rain).

In machine learning, researchers often use something called a confusion matrix to measure how reliable a model’s predictions are, although, as I will argue, there is a gulf between making accurate predictions and possessing knowledge. In essence, the confusion matrix is a table that counts how often a model gets things right or wrong. Let us say we build an AI to diagnose cancer from medical scans. The confusion matrix records how many times the AI correctly identifies cancer (true positives), correctly identifies that there is no cancer (true negatives), falsely says that there is cancer when there is not (false positives), and falsely says that there is no cancer when there is (false negatives). If an AI model consistently produces more true positives and true negatives than false ones, we consider it reliable. We can then go on to calculate metrics that summarise how well our model works overall, such as accuracy and F1 score.

At first glance, this all sounds perfectly reasonable. The AI model produces a prediction (belief), it is measured against reality (truth, which might come from expert-labelled data), and it is based on patterns in the data (justification). But here is the problem: the confusion matrix does not explain why our AI makes the predictions that it does, because it does not capture the appropriateness of its *justification*. Imagine that our AI model says a scan shows

cancer—and it is right—but it made that call based on an irrelevant smudge on the image rather than the tumour itself. In machine learning terms, this produces a true positive, a term that misleadingly implies ‘success’. Similarly, if the AI says a scan does not show cancer, and is right, that too sounds good—but what if the AI ignored key parts of the image? This would produce a true negative, again implying success, but is not genuine knowledge. These scenarios are more meaningfully described by the term *epistemic luck*—beliefs that are correct only by chance.

Philosophers have wrestled with this problem since Edmund Gettier’s famous paper in 1, in which he showed that even justified true belief is not enough to call something knowledge. Here is a simplified version of one of his examples. A man believes that his friend owns a Ford, because he recently saw him driving one. It turns out the friend really does own a Ford—but the car he was seen driving belonged to someone else. The belief was true, but only by accident. As we have seen, AI can fall into the same trap—giving us right answers by bad reasoning.

Since AI models base their predictions on data, some types of flawed human reasoning—like ignoring obvious facts and making blind guesses—cannot really happen. These include “denial” (believing something to be false that is both true and supported by evidence, such as believing that I have perfect eyesight despite a routine eye test showing otherwise), “lucky denial” (believing something to be false despite evidence suggesting it is true, but coincidentally it really was false), and “lucky guess” (believing something to be true that really is true but having no evidence, such as believing that I have correctly predicted “heads” in a random coin toss).

Setting flawed human reasoning aside, the confusion matrix still misleads and oversimplifies, because it lumps different kinds of outcomes together. For example, a belief might be held that is justified but false, such as predicting that a scan shows cancer, because there is a structure that looks malignant but is actually benign. This would be recorded as a false positive but would be mixed in with cases, where the justification turned out to be unsatisfactory,

✉ Ian van der Linde
ian.vanderlinde@aru.ac.uk

¹ Anglia Ruskin University, Cambridge, UK

such as the prediction being based on the irrelevant smudge. This is something that we could discover after the fact using explainable AI, the foundational principles for which were clearly described by Miller in 2. Similarly, a belief might be justified and true, such as predicting that a scan shows cancer when cancer truly is present *and* using relevant areas of the scan to make this prediction. This would be recorded as a true positive but would be mixed in with Gettier cases—in which the judgement was correct (scan shows cancer), but the justification turned out to be unsatisfactory, again, such as the prediction being based on the irrelevant smudge. In fact, whether the AI predicts “cancer” or “no cancer”, and whether the true answer is “cancer” or “no cancer”; in all cases, either relevant or irrelevant features from the scan could have been used to arrive at that prediction.

In real life, having good reasons for holding a belief is just as important as the belief itself. This is the basis of modern science, in which we hold beliefs that are most convincingly supported by current evidence while accepting that our beliefs may need to be revised as new evidence comes to light. This idea is known as *defeasibility*—the idea that a belief can lose its justification, and consequently its status as being true or untrue, when new evidence comes along. In contrast, confusion matrices evaluate predictions statically and do not account for the evolving nature of justification and truth over time. A belief classified as a true positive at one time might later be exposed as a false positive under deeper scrutiny. Historical ‘facts’ might need to be reconsidered with new evidence. Take the AI cancer diagnosis example. A model might be trained on data where certain patterns are labelled by experts as cancerous, but new research could later show those patterns are in fact harmless. That would mean the AI’s predictions were never really justified, even though they appeared to be at the time.

AI researchers are often preoccupied with chasing higher performance scores, but this is not enough. A model can achieve high scores, at least in part, by making predictions that are right for the wrong reasons. This can lead to overconfidence in flawed systems. If AI is going to be trusted with life-changing decisions—diagnosing disease, driving autonomous vehicles, and approving loans—we need to do better than metrics based on counts from the confusion matrix. In 3, Watson and Floridi suggested an “explanation game” that frames explainability as a discussion between

an AI and its user, rooted in philosophical ideas about justification. Here that sentiment is specifically aimed at the confusion matrix—the gold standard for describing machine learning performance that nevertheless fails to truly reveal our model’s *knowledge* in the philosophical sense. Moving forward means confronting the limitations of the confusion matrix and the metrics calculated from it and focusing on methods that allow us to expose flawed reasoning, like explainable AI. Not doing so risks building AI systems that look reliable on paper but could fail disastrously when it matters most.

Curmudgeon Corner Curmudgeon Corner is a short opinionated column on trends in technology, arts, science and society, commenting on issues of concern to the research community and wider society. Whilst the drive for super-human intelligence promotes potential benefits to wider society, it also raises deep concerns of existential risk, thereby highlighting the need for an ongoing conversation between technology and society. At the core of Curmudgeon concern is the question: What is it to be human in the age of the AI machine? -Editor.

Author contribution This is a single-author article. Could add “I.V. wrote the manuscript”, but this seems redundant.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

References

- Gettier EL (1963) Is justified true belief knowledge? *Analysis* 23(6):121–123. <https://doi.org/10.2307/3326922>
- Miller T (2019) Explanation in artificial intelligence: Insights from the social sciences. *Artif Intell* 267:1–38. <https://doi.org/10.1016/j.artint.2018.07.007>
- Watson DS, Floridi L (2021) The explanation game: a formal framework for interpretable machine learning. *Ethics, Governance, and Policies in Artificial Intelligence*. Springer International Publishing, Cham

Publisher’s Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.